EMOTICORE

See what humans miss. Detect what systems ignore.

TYPE	SCOPE	RUNTIME	VERSION
Project Presentation	**4 Pages · Full Product**	**100% On-Device**	**1.0 · Jul 2026**

THE PROBLEM

Security systems watch. They don't understand.

Cameras and microphones capture more footage than any human can meaningfully review — and by the time a threat is visually obvious, the moment to act on it has often already passed.

01

Footage without interpretation

Raw video and audio streams record everything but understand nothing. Intent, stress, and escalation are invisible to a passive feed.

02

Operators can't watch everything

One person monitoring dozens of feeds will miss the subtle signal — a raised voice, a flash of fear — buried in hours of routine footage.

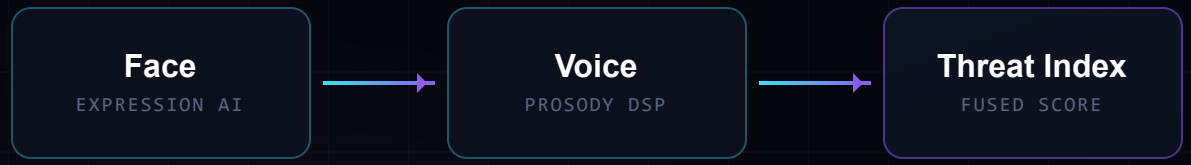
03

Detection arrives too late

By the time an incident is visually unmistakable, the window for calm de-escalation or early intervention has usually already closed.

THE SOLUTION

Fuse what the camera sees with what the voice reveals — into one live threat picture.



WHAT WE BUILT

One landing page. Three working products.

Every demo genuinely runs the AI — nothing here is a mockup.

INDEX.HTML

Landing Page

Enterprise marketing site — animated hero, capability showcase, pipeline explainer, deployments, and a simulated ops console teaser.

DEMO.HTML

Camera Demo

Real facial detection and 7-class emotion classification on your live webcam — 68 landmarks, stable per-person tracking, threat sparkline.

VOICE.HTML

Voice Demo

Record or upload a voice clip; an on-device prosody engine scores six emotions plus a vocal stress index, with saved analysis history.

CONSOLE.HTML — FLAGSHIP

Operations Console

The full product: live camera + voice fusion, session recording, an incident engine, a telemetry timeline, and exportable reports.

HOW IT WORKS

Feed to alert in under 400 milliseconds.

A continuous four-stage pipeline runs against every connected stream.

01

Capture

Camera and microphone streams via standard browser APIs — no proprietary hardware.

02

Analyze

Face and voice models run in parallel, entirely on-device.

03

Score

Signals fuse into one confidence-weighted threat index.

04

Alert

Sustained elevation opens a logged, exportable incident.

MEASURED END-TO-END · P95 **~380ms** · RUNS ENTIRELY IN-BROWSER

UNDER THE HOOD

Three neural networks. One DSP engine. One fusion formula.

FACIAL PIPELINE — NEURAL NETWORKS

tinyFaceDetector	Face detection, Tiny-YOLO style
faceLandmark68TinyNet	68-point facial landmarks
faceExpressionNet	7-class expression softmax

VOCAL PIPELINE — SIGNAL PROCESSING

Autocorrelation F0	Pitch detection, 60–400Hz
RMS / Jitter / Rate	Loudness, stability, pace

THREAT FUSION FORMULA

```
facial = weighted(angry, fearful, disgusted…)
fused = 0.65×facial + 0.35×vocal + lexicon
index ← index + (fused − index) × 0.12

LOW < 35 ≤ ELEVATED < 65 ≤ HIGH
```

All weights and thresholds are operator-tunable in the console's settings drawer — nothing is hardcoded.

THE FLAGSHIP

Inside the Operations Console

Live Fusion
Camera and voice analyzed simultaneously; mic denied falls back to face-only gracefully.

Incident Engine
Hysteresis-based detection captures a video snapshot and logs peak, duration, and driver.

Tunable Settings
Fusion balance, thresholds, and the threat lexicon are editable and saved to this browser.

Telemetry Timeline
120-second scrolling chart with threshold bands and three plotted signals.

Session Reports
One-click JSON and CSV export of the full session — generated and downloaded locally.

Command Palette
Ctrl+K opens a filterable, keyboard-driven action menu for every console function.

PRIVACY BY DESIGN

Nothing you show or say ever leaves this device.



ONE OPT-IN EXCEPTION: LIVE TRANSCRIPTION ROUTES AUDIO THROUGH THE BROWSER'S SPEECH SERVICE — OFF BY DEFAULT, LABELED WHEN ENABLED

TECHNOLOGY STACK

No framework. No build step. No server.

CORE

HTML5 · CSS3 · Vanilla **JavaScript** — zero dependencies to install

FACE AI

@vladmandic/face-api on TensorFlow.js, loaded from CDN, runs entirely client-side

AUDIO

Hand-built DSP on the **Web Audio API** — pitch, jitter, loudness, rate

TYPOGRAPHY

Sora, **IBM Plex Sans** & **Mono** — Google Fonts

STORAGE

localStorage for settings/history · Blob downloads for reports

DELIVERY

Installable **PWA** — manifest + service worker, offline-capable

NO SERVER · NO DATABASE · NO USER ACCOUNTS — THE BROWSER IS THE ENTIRE APPLICATION

DESIGN SYSTEM

Defense-tech mission control

Instrument panels, hairline borders, monospace telemetry — restraint over neon.

SORA — DISPLAY

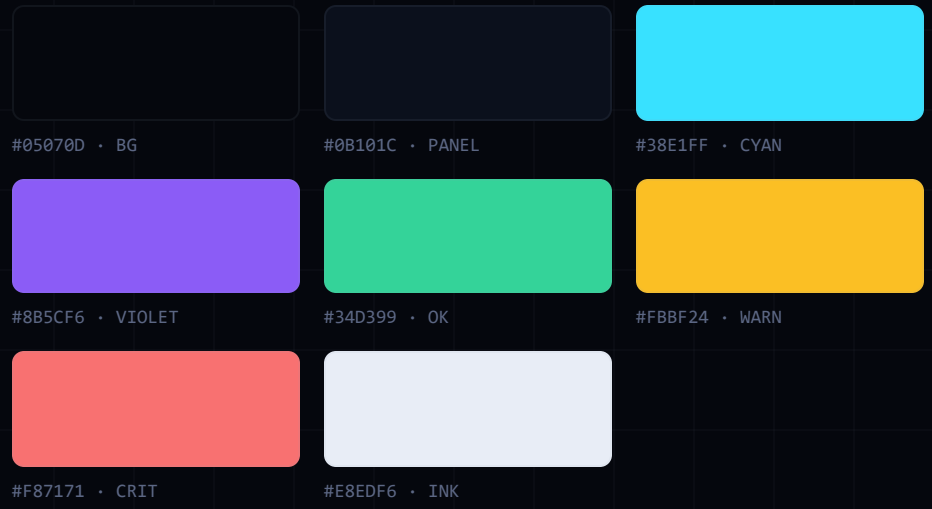
Detect intent.

IBM PLEX SANS — BODY

Behavioral intelligence for security teams.

IBM PLEX MONO — TELEMETRY

`THREAT IDX · 18 · LOW`



WHAT'S NEXT

Roadmap

01 **Deploy to a real URL**
GitHub Pages — camera, mic, and PWA install all work properly over https.

02 **Session persistence**
Auto-save sessions locally; a session browser and comparison view.

03 **Neural voice-emotion model**
Upgrade the prosody heuristic once a browser-ready speech-emotion model matures.

04 **Multi-camera console**
Several simultaneous streams feeding one combined incident rail.